
Network Swiss Army Knife (NSAK)

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The Challenge

Rising Cyber Threats

- Attackers already leverage AI — defenders must keep pace
- Red & blue team exercises demand deep expertise
- High recurring effort — inaccessible for small and medium-sized enterprises
- Scripted automation is brittle and inflexible

Our Approach



Agentic AI

- › Integrate an LLM agent into NSAK via LangChain
- › Equip it with tool calling for autonomous execution
- › Benchmark against a scripted reference scenario

Research Questions



Three Open Questions

- Can an AI agent match a scripted scenario at automated network reconnaissance?
- How effectively can AI assist a human operator during a live security investigation?
- Does it hold up in real-world networks beyond a controlled lab?

3 LLMs • 2 Network Environments • Static Reference Baseline

Network Swiss Army Knife (NSAK)

Predecessor Project

- › Modular CLI framework for network security scenario automation
- › Resources: Devices, Environments, Scenarios and Drills
- › Established in prior work — this thesis extends it with AI

What We Built



AI Reconnaissance Scenario



Static Reference Scenario



Reproducible Network Environment



Benchmark Suite

Evaluation Highlights



Containerlab (Virtual)

- ✓ All 3 models completed reconnaissance autonomously
- ✓ Multi-agent decomposition improved smaller models significantly
- Larger models showed no benefit from decomposition



BFH Network Security Lab (Physical)

- ✓ Agent navigated a live network with 35+ hosts
- Structured output failed — adapted to unstructured mode
- Small local model (qwen3:8b) could not complete the task

Static scenario: fast & reproducible, but no reasoning or report generation

Demo: Interactive AI Agent

-  **Approval workflow** — operator confirms each drill before execution
-  **AI reasoning** — model selects tools, adapts strategy, explains decisions
-  **Email report** — prioritized findings delivered as a human-readable summary

Prompt:

Help met to find all vulnerable services on interface
etho

Prompt:

Please check if the SMB share is insecure or if you can access user accounts from LDAP with weak credentials

Prompt:

Please explain what the leaked documents are revealing

Prompt:

Please check if you can enumerate all services via DNS,
maybe you can even do a full zone transfer

Prompt:

I think we gathered enough information now, create a summary in Markdown and send it as an email

Key Findings



AI Scenario

- ✓ Prioritized, natural-language reports
- ✓ Adaptive to unknown environments
- ✓ Low barrier — minimal expertise needed
- Variable output, hallucination risk
- Requires human supervision



Static Scenario

- ✓ Scripted & reproducible
- ✓ Fast, no token cost
- ✓ Precise, auditable steps
- No reasoning or report generation
- Requires domain expertise to build

“AI and scripted automation are complementary — not competing — approaches.”